

Pre-Stall Load Redistribution on Swept Tailplane Models with Full-Span Controls

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Pre-stall lift and pitching-moment changes on swept tailplane models with full-span plain controls are examined using MONNALISA wind-tunnel measurements and Reynolds-averaged Navier–Stokes solutions. The analysis distinguishes coefficient increments from departures relative to preceding incidence trends. For the positive-sweep, zero-dihedral model at a 20° deflection, computed fixed-surface lift continues increasing but grows more slowly than the preceding trend. The fixed surface accounts for about 63% of the total lift departure across the tested baseline windows and zero-incidence solutions, whereas the control contributes more strongly to the moment departure about the fitted aerodynamic center. Sectional pressures and spatial load integrations relate this difference to increased leading-edge suction, weaker hinge/aft loading, and different spanwise moment weights. Matched dihedral comparisons quantify changes in load projection and moment contributions with incidence. Agreement with measured integrated coefficients is partial, so the detailed loading interpretation concerns the computed states. Continuing lift growth can thus coexist with reduced growth of both component contributions and a different component ranking in pitching moment.

Nomenclature

b	=	undeformed model semispan, m
$c, \bar{c}$	=	local chord and mean aerodynamic chord (MAC), m
C_L, C_D	=	lift and drag coefficients
$C_{F,x}, C_{F,z}$	=	body-axis force coefficients
C_M^A	=	pitching-moment coefficient about the respective fitted center
$C_M^O, C_M^{A_I}$	=	moment about numerical origin or common Geom. I fitted axis

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$C_p, \mathbf{C}_f$	=	pressure and vector skin-friction coefficients
$D_{Q,j}$	=	departure of component j from its preceding coefficient trend
f_{rev}	=	specified upper-aft chordwise reversal area fraction
$m_{Q,j}^{\text{pre}}$	=	fitted preceding slope, per degree
$M_\infty, Re_{\bar{c}}$	=	freestream Mach and MAC Reynolds numbers
q_∞, S	=	dynamic pressure and fixed semiplanform reference area
$s, \eta = s/b$	=	undeformed span coordinate and its nondimensional value
u_i	=	supplied propagated uncertainty magnitude at sample i
x_A, x_O	=	numerical fitted-center and original reference abscissae, m
α, δ, Γ	=	displayed incidence, control deflection, and dihedral, deg
λ, Λ, τ	=	quarter-chord sweep, aspect ratio, and taper ratio
ξ	=	local straightened chord fraction
BCM, SA	=	Bas–Cakmakcioglu transition and Spalart–Allmaras closure
RANS, WT	=	Reynolds-averaged Navier–Stokes and wind tunnel

I. Introduction

Separation and reduced effectiveness at large plain-flap deflections are established features of flapped lifting surfaces [1]. On swept finite wings, planform and three-dimensional flow also influence load distribution and stall progression [2]. Full-span plain controls have been studied on finite wings: Campbell, Blanks, and Leaver [3] measured lift, drag, and pitching moment on low-aspect-ratio rectangular models with several flap chords.

The aerodynamic question considered here is how local loading changes combine in the integrated pre-stall response. A component can continue gaining lift while falling below its preceding growth trend. Comparing ordinary increments with departures from that trend quantifies this distinction. Pitching moment additionally weights where the changes occur: similar force changes at different chordwise or spanwise locations can produce different moment contributions.

The MONNALISA database publication [4], Sec. IV and Fig. 11, documents the baseline lift nonlinearity, expanding control-surface reversed flow, pressure changes, and continued growth of total lift, as well as nonlinear lower-fidelity behavior. The present study extends that account by quantifying how changes on the fixed and control surfaces combine in lift and pitching moment, where the changes occur, and how they differ with deflection and dihedral. Measured lift and moment responses are characterized separately to establish the experimental context for the numerical loading interpretation.

The test articles are isolated tailplane-inspired wings, without a fuselage, wing wake, or complete-aircraft trim condition. Their full-span, 30%-chord controls define the tested extent; partial-span controls and other flap chords are outside the dataset. The 45° dihedral setting provides a large change in orientation on the same planform. It is a strong geometric contrast, rather than a typical conventional-tail setting. A second planform changes several geometric parameters together.

Nonlinear lifting-line methods can incorporate sectional viscous data [5]. In MONNALISA, the DUST lifting-line formulation uses sectional tables and corrections and gives nonlinear responses, while the retained potential-based panel and vortex-lattice predictions are smoother [4]. The available behavior is therefore model-specific. The RANS solutions provide the surface loads needed for the present analysis; their integrated lift and moment are first assessed against the measurements.

II. Experimental, Numerical, and Analysis Framework

A. Models and experimental installation

The models have an undeformed semispan $b = 1.6$ m, nominal thickness ratio $t/c = 9\%$, and a full-span plain trailing-edge control rotating about the 70%-chord hinge. Table 1 gives the geometry and global references. Aspect ratio is defined for the corresponding full planform, $\Lambda = 2b^2/S$. Geom. I at $\Gamma = 0^\circ$ and 45° retains the same undeformed planform, area, and MAC. Geom. II simultaneously changes sweep, taper, aspect ratio, area, and chord distribution. It is a planform comparison, rather than an isolated sweep-direction test.

Table 1 Geometry and reference quantities. Area is the undeformed semiplanform area; $Re_{\bar{c}} = Re_b \bar{c}/b$, with $Re_b = 5.5 \times 10^6$.

Property	I, $\Gamma = 0^\circ$	I, $\Gamma = 45^\circ$	II, $\Gamma = 0^\circ$
λ (deg)	30	30	-15
Λ	5.0	5.0	5.2
τ	0.4	0.4	0.6
S (m ²)	1.024	1.024	0.984604
$\bar{c}$ (m)	0.679184	0.679184	0.6282
Root chord (m, rounded)	0.914286	0.914286	0.76923
$Re_{\bar{c}}$	2.335×10^6	2.335×10^6	2.159×10^6
Numerical x_A (m)	0.642000	0.643532	-0.003618

The WT measurements were obtained in the low-turbulence section of the closed-circuit GVPM facility at Politecnico di Milano. Figure 1 shows the floor-mounted semispan installation. The dummy floor conceals the support, balance, and control actuation mechanism. The documented test-section width and length are 4 and 6 m, respectively; the undeformed semispan is 40% of that width. A consistent installed-height value and a floor-layer profile at the model are unavailable for a quantitative confinement assessment. Thus neither a thin floor layer nor negligible wall effects is assumed.

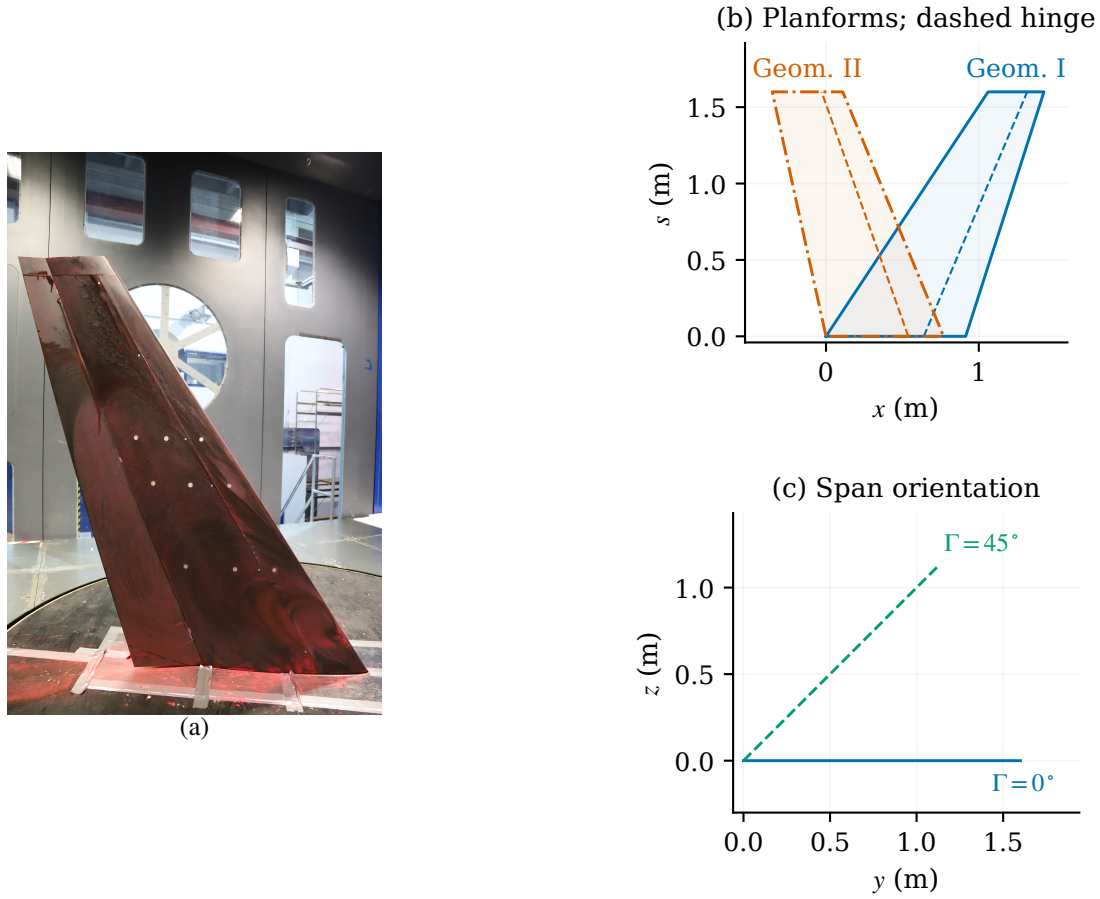


Fig. 1 Model installation and geometry: (a) oil-flow photograph at $\alpha = 25^\circ$, $\delta = 20^\circ$, illustrating the GVPM mounting rather than the pre-stall flow state; (b) undeformed planforms and 70%-chord hinges; (c) metric span orientation in numerical axes, with $b = 1.6$ m at both dihedral settings. Dimensions are given in Table 1.

The database setup description [4] gives freestream turbulence below 0.1%, measured using a turbulence sphere and hot-wire anemometry. Outside the wall layers, the inlet total-pressure variation was within $\pm 0.5\%$ of dynamic pressure over most of the section and within $\pm 1\%$ overall; outlet measurements confirmed similar uniformity. The reported longitudinal static-pressure gradient is $\partial C_P / \partial x = -0.004 \text{ m}^{-1}$. Loads were acquired through a RUAG 767 strain-gauge balance below the floor and an HBM QuantumX system; the reduction includes the balance-to-wind transformation and the database corrections. Nominal conditions are $M_\infty = 0.147$, $T_\infty = 300 \text{ K}$, and the MAC Reynolds numbers in Table 1. The span-based database Reynolds number is retained only for traceability.

The measurements use the LE1 profile variant. Recorded incidences are $-25, -24, -22, \dots, 22, 24, 25^\circ$: the local pre-stall spacing is 2° , with 1° endpoint steps. The comparisons use $\delta = 0, 10, 20, 30^\circ$. Uncertainty bars are the supplied propagated measurement/reduction-chain magnitudes, including balance and transformation contributions. Each fitted-center moment retains its associated propagated uncertainty. A common coverage factor and between-incidence covariance are unavailable; the experimental contrast sensitivities below state their assumptions explicitly.

B. Conventions and moment references

In the numerical body axes, x is downstream, y is spanwise, and z is upward. The force rotation and normalization are

$$C_L = C_{F,z} \cos \alpha - C_{F,x} \sin \alpha, \quad C_D = C_{F,z} \sin \alpha + C_{F,x} \cos \alpha, \quad C_{F,k} = F_k / (q_\infty S). \quad (1)$$

Pitching moment is positive nose-up and normalized by $q_\infty S \bar{c}$. Table 2 summarizes the retained angle and frame mapping. No further angle flip or alignment shift is applied to improve agreement.

Table 2 Conversion to the displayed convention. Subscript r denotes the existing reduced WT record; n denotes a native numerical state.

Source	Displayed quantities and state handling
Reduced WT	$\alpha = -\alpha_r$, $\delta = \delta_r$, $C_L = -C_{L,r}$, $C_M^A = -C_{M,r}^A$; drag and uncertainty magnitudes retain their signs.
RANS, direct	Native α_n , δ_n retained; forces use Eq. 1.
RANS, $\Gamma = 0^\circ$ reflected	Opposite-deflection parent gives $\alpha = -\alpha_n$, $\delta = -\delta_n$. Positions/forces transform with $T = \text{diag}(1, 1, -1)$; moments with $(\det T)T$; physical sides exchange.
RANS, $\Gamma = 45^\circ$	Signed incidence is directly computed; no same-dihedral reflection.
Zero incidence	Direct and symmetry-reflected zero-incidence solutions remain distinct. Each field uses the force state of its corresponding solution.

The principal WT–RANS comparison uses moments about the respective zero-deflection fitted aerodynamic centers, denoted C_M^A . Following the database construction [4], the reference is obtained from linear fits to zero-deflection moment and force over $|\alpha| \leq 10^\circ$ and then held fixed for all control deflections. The WT reduction uses uncertainty-weighted fits; the retained numerical references use their historical linear fits. This compares the same reference construction across datasets, rather than identical physical axes. It is neither an instantaneous aerodynamic center nor necessarily quarter-MAC. The WT moment is not transported onto a numerical axis without a physical registration.

The original numerical reference is $O = (0.228571, 0, 0)$ m, the root quarter-chord of Geom. I. For a numerical pole displaced only in x ,

$$C_M(x_1) = C_M(x_0) + \frac{x_1 - x_0}{\bar{c}} C_{F,z}. \quad (2)$$

The fitted abscissae are in Table 1. Matched dihedral diagnostics use the identical numerical axis $A_I = (0.642000, 0, 0)$ m in both configurations, denoted $C_M^{A_I}$. Geom. II uses its own MAC once for every moment contribution; all coefficients use the fixed areas in Table 1.

C. Numerical method and existing accuracy evidence

The existing SU2 [6] solutions use compressible ideal-gas RANS at the nominal WT Mach and Reynolds conditions, with Sutherland viscosity. Unstructured finite volumes use median-dual vertex control volumes. Mean-flow convection

uses Roe fluxes with second-order MUSCL reconstruction [7], Green–Gauss gradients, and a Venkatakrishnan limiter [8] with coefficient 0.05. SA [9] convection uses scalar upwinding without MUSCL, hence first-order reconstruction. The BCM transition closure [10] supplies an algebraic intermittency correction, not an independently transported transition equation. Spatial order must therefore be distinguished between mean-flow and SA equations.

The surfaces have no-slip, zero-heat-flux boundaries; the exterior uses a far-field boundary and the root plane a symmetry boundary. This excludes the WT floor, its layer, and the support. Implicit Euler pseudo-time advancement, FGMRES linear solves, adaptive CFL, and staged restarts were used to obtain steady states. The configured Cauchy stopping criterion monitors lift and drag over 100 entries, commonly at 3×10^{-6} and at 10^{-6} in later Geom. II stages. Moment is not a stopping quantity; iteration caps and restarts also occur.

Terminal histories match the force reports for 41 of the 44 interpreted surface states; state-specific exceptions and incomplete final-stage registration are detailed in Appendix A. For the inclined $\delta = 20^\circ$ irregularity over $-6 \rightarrow -4^\circ$, last-100 lift ranges are approximately 10^{-4} , much smaller than the between-state change. These iterative checks complement the mesh and experiment–model comparisons.

The automated mesh procedure [4] uses anisotropic surface triangles, leading-edge and hinge/junction refinement, a hex-dominant octree volume, and incidence-oriented wake refinement. Nominal controls are a 2.5° leading-edge angle, 1° junction angle, seven trailing-edge points, maximum surface spacing 0.02 m, and boundary-layer growth ratio 1.15. Actual first-cell height and layer count are not registered for every retained mesh. Representative stored y^+ medians are 0.26 for $I/\delta = 20^\circ/\alpha = 2^\circ$, 0.19 for $I/\Gamma = 45^\circ/\delta = 30^\circ/\alpha = 0^\circ$, and 0.23 for $II/\delta = 20^\circ/\alpha = 0^\circ$, with maxima approximately 1.22, 1.26, and 1.13. These support resolved near-wall treatment for those samples.

Selected surface meshes have 122,860 points/182,423 elements for $I/\delta = 20^\circ$ at 0, 2, 4°, 141,828/209,641 for $I/\Gamma = 45^\circ/\delta = 30^\circ$ at $-2, 0, 2^\circ$, and 105,506/156,333 for $II/\delta = 20^\circ$ at 0, 2, 4°. These are surface counts. The retained volume for the $I/\alpha = 0^\circ/\delta = -20^\circ$ parent contains 9,440,030 points and 18,918,212 volume elements; equivalent registered volume counts are not available for every interpreted state.

Table 3 gives grid-size ranges for the existing four-level study, and Fig. 2 shows its coefficient behavior. It concerns Geom. I at $\delta = 35^\circ$, $\Gamma = 0, 45^\circ$, and $\alpha = 0, 18^\circ$. The published lift/drag assessment is augmented only by moments linked to the same retained states. Nominal-to-finest differences are 1.20–2.14% for lift and 1.13–2.49% for drag. The moment gap at $\Gamma = 0^\circ$, $\alpha = 18^\circ$ is 0.01343; the inclined sequence is nonmonotonic. This is inherited integrated-load sensitivity evidence, without an asymptotic-order or discretization-uncertainty estimate. It does not establish convergence of the current 20/30° departures, regional loads, pressure sections, or separation topology, nor of Geom. II.

D. Additive loading and empirical departures

Solver surface groups define the fixed surface (upper/lower fixed panels), the control (upper/lower control panels, hinge junction, and blunt trailing edge), and tip. Pressure and viscous contributions are retained on both physical sides

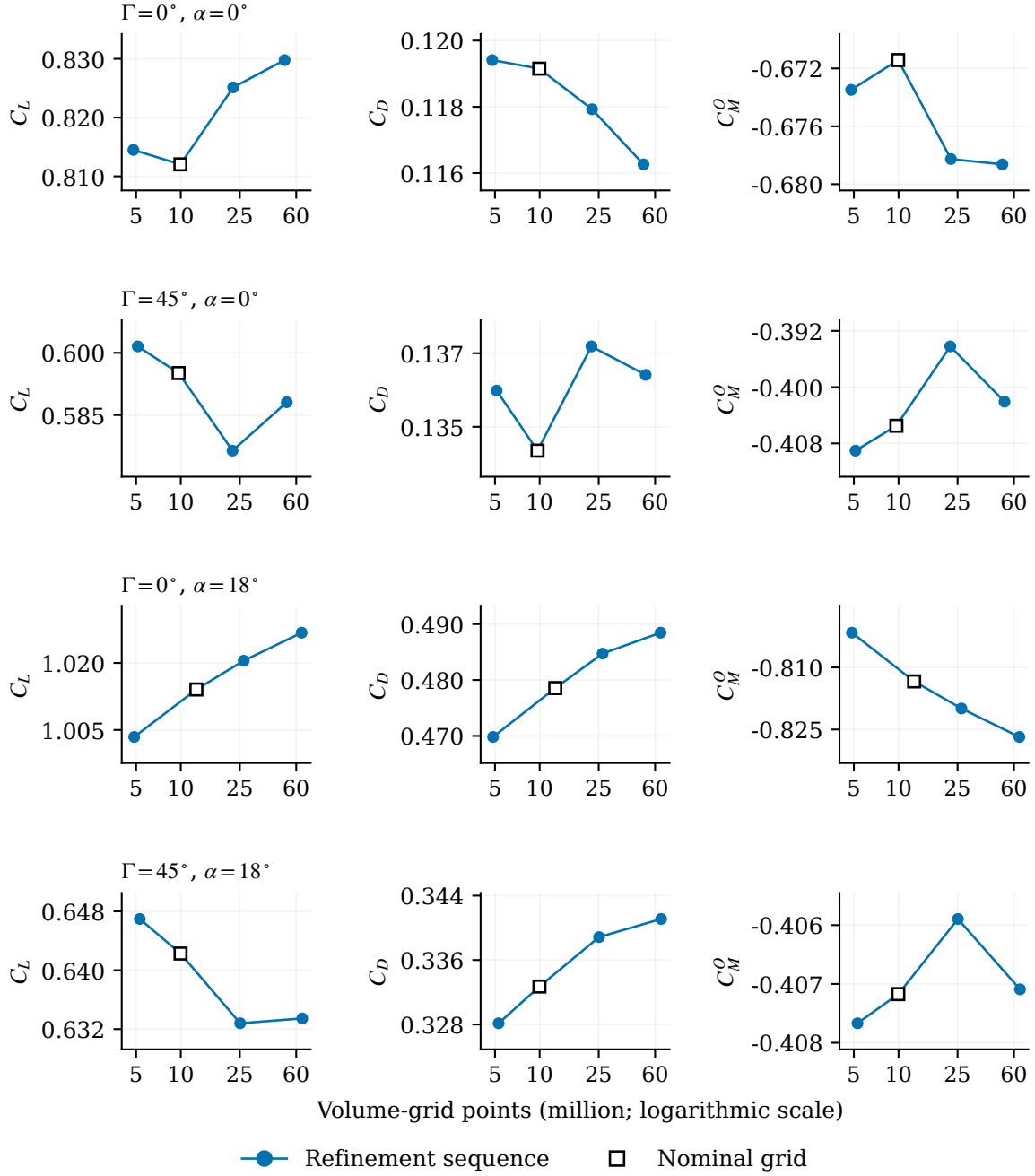


Fig. 2 Existing four-level Geom. I study at $\delta = 35^\circ$: rows show $(\Gamma, \alpha) = (0, 0), (45, 0), (0, 18), (45, 18)^\circ$; columns show C_L , C_D , and C_M^O . The abscissa is the actual volume-grid point count on a logarithmic scale; left-to-right samples are coarse, nominal, fine, and very fine. Open squares identify nominal grids. Moments use $O = (0.229, 0, 0)$ m and the Geom. I area/MAC. The two finest $\alpha = 0^\circ$ moments are taken from the same terminal histories as lift and drag; the other moments use matching force reports. Volume-element counts range from 9.51 to 109.98 million. Lines connect retained refinement levels, without an asymptotic-convergence estimate.

Table 3 Grid-size ranges of the existing four-level study. Counts are volume-grid points and volume elements, in millions. Each range spans Geom. I at $\Gamma = 0, 45^\circ$, $\alpha = 0, 18^\circ$, and $\delta = 35^\circ$; Geom. II is not part of this study.

Refinement	Points	Volume elements
Coarse	4.79–5.29	9.51–10.90
Nominal	9.67–12.74	19.06–24.07
Fine	22.33–26.56	39.93–45.70
Very fine	50.11–65.89	87.35–109.98

with the same global S and $\bar{c}$. Tip is included in every total; no component is multiplied for display.

For coefficient Q and component j , the ordinary increment is the change between two sampled incidences. Its departure from the preceding trend subtracts the increment predicted by continuation of that trend:

$$\Delta C_{Q,j} = C_{Q,j}(\alpha_b) - C_{Q,j}(\alpha_a), \quad D_{Q,j} = \Delta C_{Q,j} - m_{Q,j}^{\text{pre}}(\alpha_b - \alpha_a). \quad (3)$$

The preceding slope is an unweighted linear least-squares fit to the declared actual samples. All additive components and the total use identical samples and weights, preserving their sum. The primary $\delta = 20^\circ$ baseline is $\{-6, -4, -2\}^\circ$; the early $I/\delta = 30^\circ$ numerical baseline is $\{-16, -14, -12\}^\circ$. Each is an empirical reference, without an assumption that the preceding flow is attached. Adjacent windows and the direct and symmetry-reflected zero-incidence solutions test sensitivity. Curved sequences are described using signed finite differences rather than a forced unique event amplitude.

For WT contrasts, Eq. 3 is written as $D = \sum_i w_i C_i$, combining repeated uses of any sample into one weight. The two reported sensitivities are

$$u_{\text{ind}} = \left[\sum_i (w_i u_i)^2 \right]^{1/2}, \quad u_{\text{L1}} = \sum_i |w_i| u_i. \quad (4)$$

The first is conditional on independence; the second bounds a perturbation when each sample varies within its supplied magnitude. Neither is an established confidence interval or a new experimental uncertainty budget. Covariance-dependent significance is not inferred.

Saved nodal C_p and vector $\mathbf{C}_f$ are integrated over the complete triangulated surface. Linear traction and position-times-traction are integrated exactly within each triangle, with

$$C_M(x_A) = \frac{1}{q_\infty S \bar{c}} \int [z dF_x - (x - x_A) dF_z]. \quad (5)$$

The two terms are retained separately where a common-axis dihedral comparison needs them. All 44 full-surface reconstructions have endpoint residuals below 1.5×10^{-4} . Maximum lift and moment residuals are 2.71×10^{-5} and 7.06×10^{-5} . Conservative absolute-weight residual bounds are checked separately against each increment or departure, with a 10% effect-scale limit for an adopted interpretation. For the baseline lift departure, the bound is 1.01×10^{-5}

against $|D_L| = 0.11060$. Small shear changes that fail their corresponding bound are not assigned a resolved mechanism. Accounting closure and reconstruction agreement are distinct from iteration, mesh sensitivity, and experiment–model discrepancy; global residuals do not bound each regional error.

Geometric bins use $\eta = s/b$ and straightened ξ , obtained by undoing the control rotation about the three-dimensional hinge. They include both sides and root extensions; end faces are retained separately. Whole triangles are assigned by centroid. These geometric regions differ from the solver surface groups, especially in their assignment of the hinge junction. Sectional pressure curves use $\eta = 0.2, 0.5, 0.8$ and preserve actual contour connectivity on the native upper/lower fixed and control panels, excluding junction and end faces. Three sections illustrate pressure changes; complete integrations supply the budgets.

A chordwise reversal indicator measures the actual upper geometric-aft area ($\xi \geq 0.7$, excluding end caps) on which $\mathbf{C}_f \cdot \mathbf{t}_c < -\epsilon$, with downstream chord direction projected into the face tangent. Linear nodal signs are clipped within triangles. The fraction f_{rev} is tested at $\epsilon = 0, 10^{-6}, 10^{-5}$. It measures a specified component of reversed shear, rather than the full three-dimensional separation footprint.

III. Measured Response and Numerical Correspondence

Figure 3 presents the baseline measured polars, including the lower 10° deflection. Figure 4 compares the three configurations at their respective fitted moment references. Its separate strips show selected WT incidence intervals, whose resolution and uncertainty are assessed below. Table 4 gives selected preceding-trend contrasts and their uncertainty sensitivities.

Table 4 Primary LE1 WT departure contrasts. Each baseline is $\{-6, -4, -2\}^\circ$; brackets are selected separately for lift and moment. Moment uses the WT fitted center. u_{ind} and u_{L1} are the full-contrast sensitivities of Eq. 4, not confidence limits.

Geom., Γ, δ	Coefficient	Interval (deg)	D	u_{ind}	u_{L1}
I, $0^\circ, 20^\circ$	C_L	$-2 \rightarrow 0$	-0.08274	0.02357	0.03631
I, $0^\circ, 20^\circ$	C_M^A	$-2 \rightarrow 0$	+0.03148	0.00862	0.01360
I, $45^\circ, 20^\circ$	C_L	$-2 \rightarrow 0$	-0.04785	0.01694	0.02634
I, $45^\circ, 20^\circ$	C_M^A	$-2 \rightarrow 0$	+0.00847	0.02630	0.04082
I, $45^\circ, 30^\circ$	C_L	$-2 \rightarrow 0$	-0.04038	0.01646	0.02996
I, $45^\circ, 30^\circ$	C_M^A	$-4 \rightarrow -2$	-0.02860	0.02478	0.03822
II, $0^\circ, 20^\circ$	C_L	$0 \rightarrow 2$	-0.06902	0.02090	0.03739
II, $0^\circ, 20^\circ$	C_M^A	$0 \rightarrow 2$	+0.01668	0.00793	0.01424

For $I/\Gamma = 0^\circ/\delta = 20^\circ$, the measured lift slope decreases sharply over $-2 \rightarrow 0^\circ$: lift grows by 0.0199, compared with 0.0892 over $-4 \rightarrow -2^\circ$. Over the same interval, moment increases by 0.0360, becoming less negative while retaining its sign. Both departures exceed the stated uncertainty sensitivities (Table 4); the corresponding pointwise lift- and moment-coefficient uncertainty magnitudes are approximately 0.0127–0.0133 and 0.0046–0.0047. These sampled

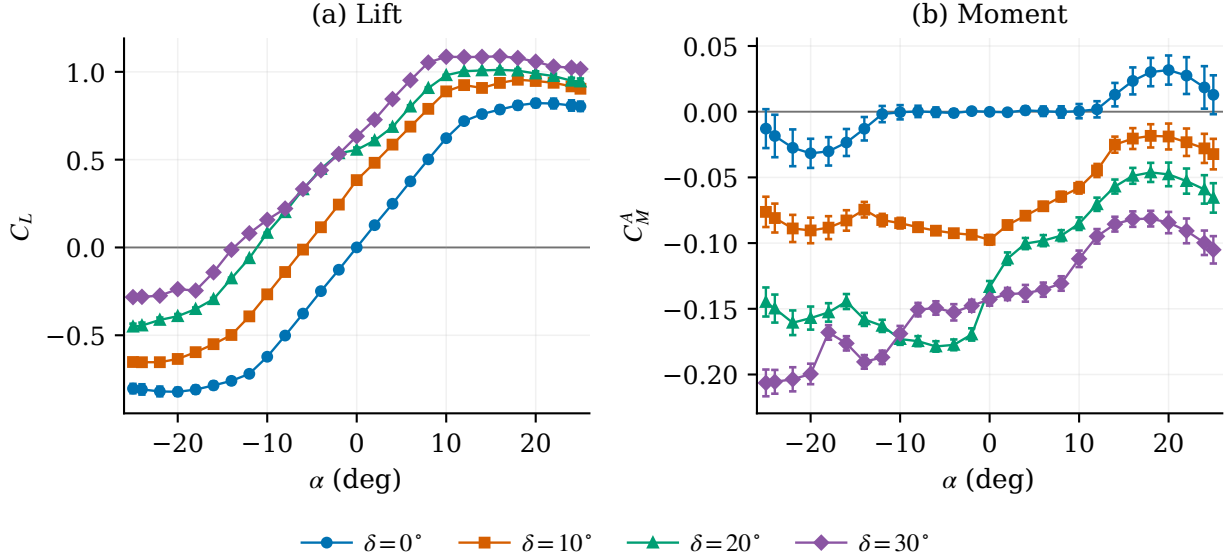


Fig. 3 LE1 WT response of Geom. I, $\Gamma = 0^\circ$, for $\delta = 0, 10, 20, 30^\circ$. Moment C_M^A uses the experimental zero-deflection fitted center, held fixed across deflection. Bars are the paired supplied propagated magnitudes, without an assigned common coverage factor.

contrasts identify the response change at the available incidence spacing.

At $\delta = 30^\circ$, the early measured lift increments are 0.0765, 0.0642, and 0.1113 over $-12 \rightarrow -10$, $-10 \rightarrow -8$, and $-8 \rightarrow -6^\circ$. Moment already increases by about 0.0180 and 0.0183 on the first two intervals. Curvature in the preceding lift sequence makes its departure window-sensitive, spanning both signs; local increments describe this broad feature more directly. At the lower 10° deflection, the moment increment changes from -0.0038 over $-2 \rightarrow 0^\circ$ to $+0.0113$ over $0 \rightarrow 2^\circ$. Thus the measured moment response changes at a lower deflection than the pronounced 20° lift slowdown.

For $I/\Gamma = 45^\circ/\delta = 20^\circ$, measured lift growth remains suppressed across $-2 \rightarrow 2^\circ$. The moment increment over $-2 \rightarrow 0^\circ$ is positive, but its departure is smaller than both uncertainty sensitivities. At $\delta = 30^\circ$, the selected lift slowdown occurs over $-2 \rightarrow 0^\circ$, whereas moment decreases earlier, by 0.0265 over $-4 \rightarrow -2^\circ$. That moment departure is comparable with u_{ind} and smaller than u_{L1} . The lift and moment intervals therefore require separate uncertainty-qualified descriptions.

For $II/\delta = 20^\circ$, measured lift grows by 0.0440 over $0 \rightarrow 2^\circ$, compared with approximately 0.0921 on the preceding and following intervals. Moment increases by 0.0187. Its primary departure exceeds both sensitivities, although adjacent-window departures do not uniformly exceed the primary L1 magnitude. The 30° response is also shown in Fig. 4 for context.

The numerical correspondence differs between coefficients. For $I/\delta = 20^\circ$, the RANS lift slowdown occurs over $0 \rightarrow 2^\circ$, one sampled interval later than in WT. At zero incidence the direct and reflected numerical lift coefficients are approximately 0.8422 and 0.8434, compared with WT 0.5560 and uncertainty magnitude 0.0133. This substantial bias

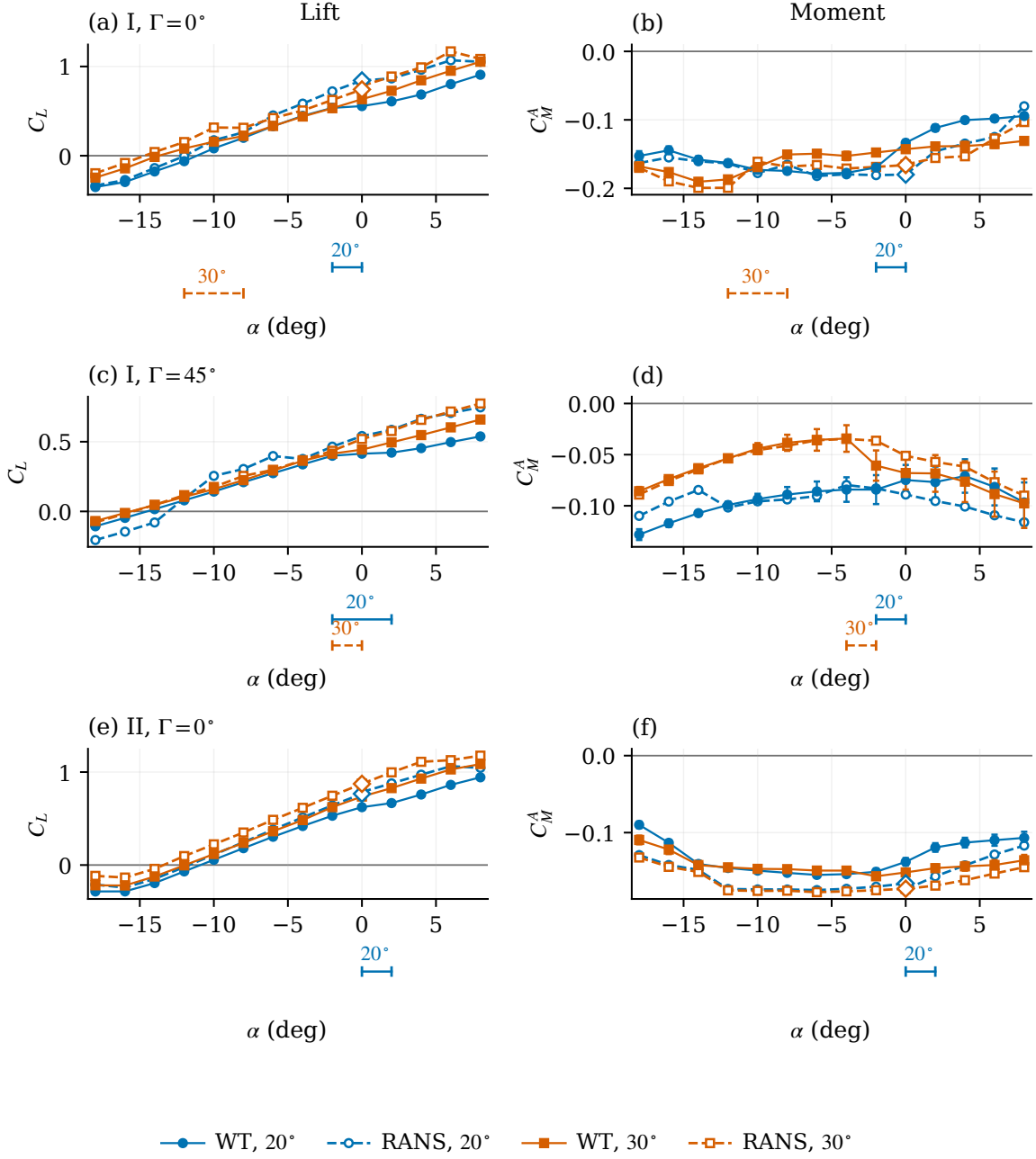


Fig. 4 WT-RANS comparison of lift and moment for $I/\Gamma = 0^\circ$ (a,b), $I/\Gamma = 45^\circ$ (c,d), and $II/\Gamma = 0^\circ$ (e,f). Filled/solid: LE1 WT; open/dashed: RANS. Circles and squares distinguish $\delta = 20^\circ$ and 30° . Separate strips show selected WT intervals, subject to the uncertainty discussion. Open diamonds identify symmetry-reflected zero-incidence solutions. Moment uses each dataset's zero-deflection fitted center; direct and reflected branches remain separate.

accompanies a qualitatively similar slowdown and less-negative moment response. The early numerical 30° lift change is sharper than the broad measured feature, and its fitted-center moment departure is window-sensitive.

At $\Gamma = 45^\circ$, numerical $\delta = 20^\circ$ lift decreases over $-6 \rightarrow -4^\circ$ while WT increases; the computed sequence does not reproduce the sustained measured slowdown or weak moment response convincingly. At $\delta = 30^\circ$, a negative moment change is present, with the strongest numerical bracket later than the measured $-4 \rightarrow -2^\circ$ interval. For $\Pi/\delta = 20^\circ$, the direct numerical $0 \rightarrow 2^\circ$ lift increment is 0.0935, more than twice WT 0.0440. The corresponding fitted-center moment increment, 0.0168, is locally close to WT 0.0187, retaining the distinct reference constructions.

This partial correspondence bounds the interpretation: the pressure, reversal, and component analyses describe the computed load response. The symmetry-root/floor difference is a modeling limitation, but the integrated comparisons alone do not assign the discrepancies to tunnel effects or validate a measured local flow topology.

IV. Joint Numerical Loading Response

A. Ordinary growth and reduced growth of the fixed surface

The reference RANS sequence shows how continuing lift growth can coexist with a substantial departure from the preceding trend (Fig. 5 and Table 5). For $I/\Gamma = 0^\circ/\delta = 20^\circ$ over direct $0 \rightarrow 2^\circ$, the fixed surface gains lift while the control loses lift, leaving a small positive total increment. The fixed-surface increase, however, is smaller than predicted by continuation of its preceding slope. Relative to the shared baseline $\{-6, -4, -2\}^\circ$, both components have negative departures, with total $D_L = -0.1106$. The fixed surface therefore contributes to the lift-growth deficit even while its own lift increases.

Table 5 Numerical ordinary increments and departures, including tip, for zero-dihedral models. Moments use each geometry’s fitted aerodynamic center (Table 1). Direct zero-incidence solutions are used. Primary baselines: $\{-6, -4, -2\}^\circ$ for $\delta = 20^\circ$, $\{-16, -14, -12\}^\circ$ for the early $I/\delta = 30^\circ$ interval. Six-decimal rounding can affect the displayed sum.

Geom., δ , interval	Measure	Fixed	Control	Tip	Total
I, 20° , $0 \rightarrow 2^\circ$	ΔC_L	+0.059287	-0.035104	-0.000004	+0.024179
I, 20° , $0 \rightarrow 2^\circ$	D_L	-0.069179	-0.041428	+0.000004	-0.110605
I, 20° , $0 \rightarrow 2^\circ$	ΔC_M^A	+0.015562	+0.018314	+0.000002	+0.033878
I, 20° , $0 \rightarrow 2^\circ$	D_M^A	+0.010689	+0.022553	-0.000004	+0.033238
I, 30° , $-10 \rightarrow -8^\circ$	ΔC_L	+0.040586	-0.043560	-0.000002	-0.002974
I, 30° , $-10 \rightarrow -8^\circ$	D_L	-0.067327	-0.052888	-0.000002	-0.120216
I, 30° , $-10 \rightarrow -8^\circ$	ΔC_M^A	-0.014959	+0.007907	+0.000002	-0.007050
I, 30° , $-10 \rightarrow -8^\circ$	D_M^A	-0.017339	+0.015008	+0.000002	-0.002328
II, 20° , $0 \rightarrow 2^\circ$	ΔC_L	+0.101449	-0.007930	-0.000010	+0.093509
II, 20° , $0 \rightarrow 2^\circ$	D_L	-0.026344	-0.010408	-0.000003	-0.036755
II, 20° , $0 \rightarrow 2^\circ$	ΔC_M^A	+0.011164	+0.005658	-0.000002	+0.016820
II, 20° , $0 \rightarrow 2^\circ$	D_M^A	+0.007512	+0.007105	-0.000001	+0.014614

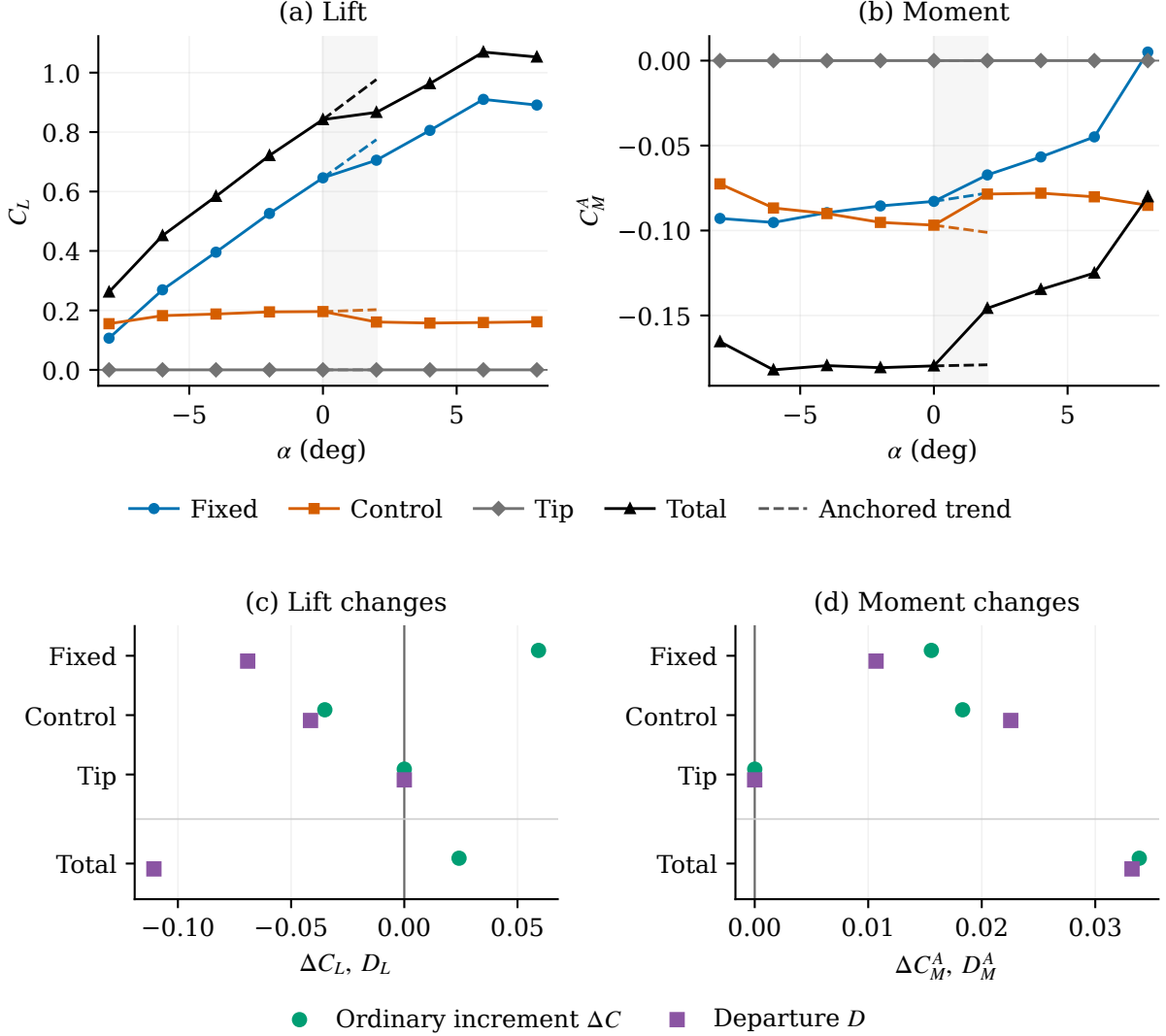


Fig. 5 $I/\Gamma = 0^\circ/\delta = 20^\circ$: component coefficients (a,b) and signed ordinary-increment/departure pairs (c,d). Dashed segments continue the fitted preceding slopes from the direct zero-incidence values; their endpoint gaps equal the departures in Eq. 3. Slopes use $\{-6, -4, -2\}^\circ$; shading marks direct $0 \rightarrow 2^\circ$. All components, including tip, use global area/MAC and fitted axis $x_A = 0.642$ m. Total is separated from the component rows; markers represent signed contributions, not uncertainty intervals.

Across three tested windows and the direct and symmetry-reflected zero-incidence solutions, the fixed fraction of total lift departure is 62.4–63.0%. The direct-state total varies from -0.1368 to -0.1050 , reflecting appreciable curvature in the earliest baseline. The allocation is more stable than the absolute amplitude. About 63% is consequently an allocation relative to these empirical trends, rather than an experimental component fraction or a causal share.

The component ranking reverses when moment is considered about the fitted aerodynamic center: both moment departures are positive, and the control contributes more strongly (Table 5). The fixed surface thus supplies the larger lift-growth deficit, whereas the control supplies the larger moment departure. Their relative importance depends on the

integrated quantity and on whether increments or departures are compared.

B. Pressure redistribution and spatial moment weighting

In the computed pressure field, upper leading-edge suction strengthens while the suction minimum near the hinge weakens at all three sampled span stations (Fig. 6). The sections cover both physical sides through the same $0 \rightarrow 2^\circ$ interval. These local loading increases and losses coexist; their contributions to the integrated response are quantified by the spatial load sums.

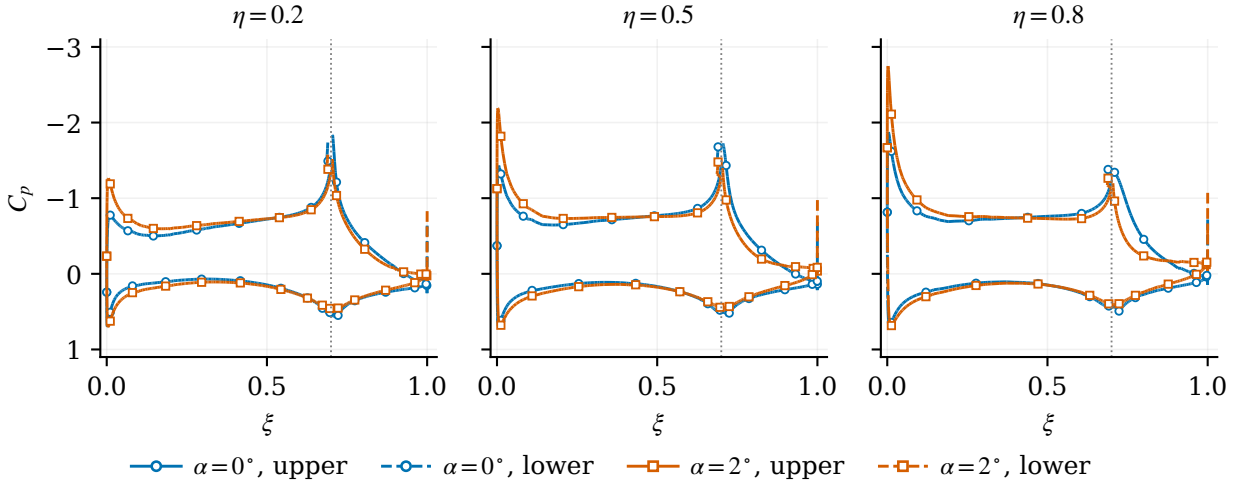


Fig. 6 Pressure sections for $I/\Gamma = 0^\circ/\delta = 20^\circ$ at direct $\alpha = 0, 2^\circ$ and undeformed $\eta = s/b = 0.2, 0.5, 0.8$. Circles/squares distinguish incidence; solid/dashed lines distinguish upper/lower sides. Sections follow the upper/lower fixed and control panels, with connectivity and face exclusions defined in Sec. II.D. The vertical dotted line marks the 70%-chord hinge. ξ is the local chord fraction obtained by undoing the hinge rotation.

The forward region gains load more slowly than its preceding trend, whereas the intermediate region loses load (Fig. 7). Together, these changes reduce the expected growth of the fixed-surface contribution. The aft region also loses lift, and all three geometric regions have negative lift departures, including both regions forward of the hinge.

Geometric fore/aft increments differ slightly from the solver fixed/control increments because the hinge junction is assigned differently. In each partition, the sum of the integrated contributions includes the entire surface. Pressure supplies nearly all the principal lift departure; the small shear contribution is retained in the total, with effect-scale limits stated in Sec. II.D.

Similar aft lift losses produce larger moment changes outboard: the moment increments about the fitted aerodynamic center increase from about 0.0025 inboard to 0.0091 outboard. This spatial weighting distinguishes the moment response from the force response. Reference and partition sensitivities are given in Appendix A; geometric region contributions are integrated quantities, not a continuous load density.

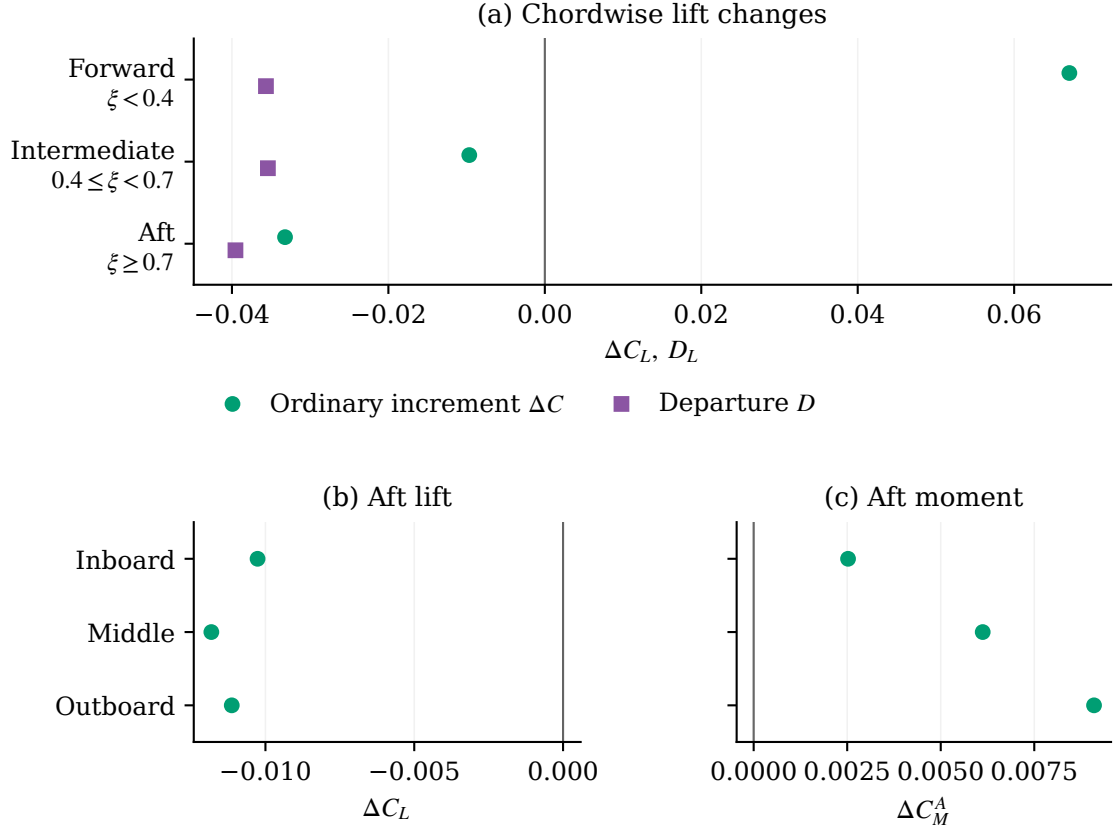


Fig. 7 Integrated geometric-region changes for $I/\Gamma = 0^\circ/\delta = 20^\circ$, direct $0 \rightarrow 2^\circ$: (a) paired chordwise lift increments and departures; (b,c) aligned ordinary aft lift and moment increments over inboard ($\eta < 1/3$), middle ($1/3 \leq \eta < 2/3$), and outboard ($\eta \geq 2/3$) regions. Baseline: $\{-6, -4, -2\}^\circ$; fitted axis: $x_A = 0.642$ m. Geometric regions follow Sec. II.D; separately retained endcaps complete the total beyond the displayed rows. Markers represent integrated contributions, rather than densities or solver surface groups.

C. Deflection contrast and reversal evolution

The early $I/\delta = 30^\circ$ interval in Table 5 also combines positive ordinary fixed lift growth with negative fixed and control departures. Its fitted-center moment contributions compete, and the total departure changes sign across tested windows. The two deflections share the lift-budget distinction, while their moment response depends more strongly on the empirical reference trend.

Figure 8 quantifies the numerical chordwise reversal accompanying these load changes. For $I/\delta = 20^\circ$, the upper-aft fraction grows from 0.146 to 0.485 over direct $0 \rightarrow 2^\circ$, with reversal already present at -2° . For $\delta = 30^\circ$, it grows from 0.622 to 0.935 over $-10 \rightarrow -8^\circ$. Expansion accompanies reduced aft loading and the pressure-pattern change. Here, pressure recovery denotes the chordwise rise downstream of the hinge suction minimum; it does not imply an attached boundary layer.

For $I/\Gamma = 45^\circ/\delta = 30^\circ$, the control is already extensively reversed over the selected moment interval. Later inclined

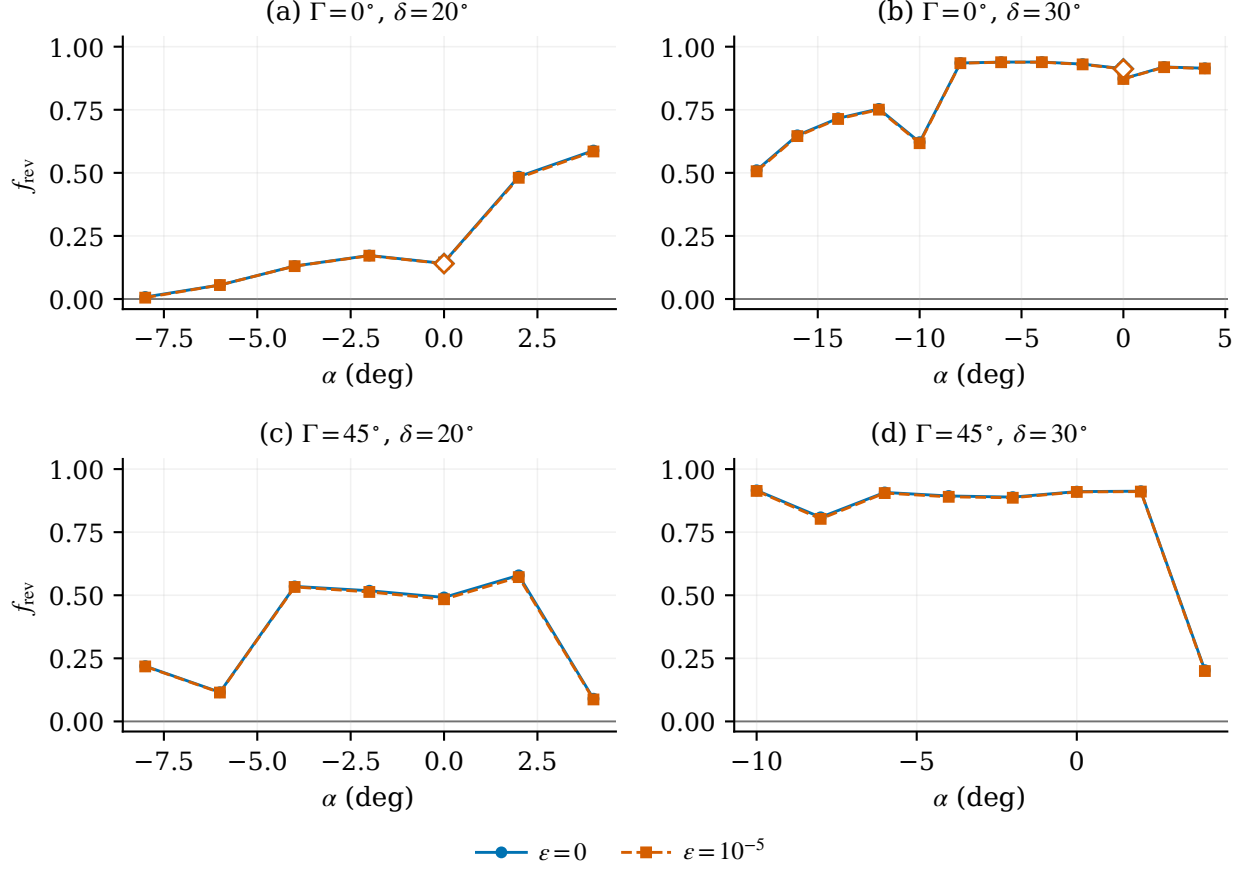


Fig. 8 Upper geometric-aft chordwise reversal fractions for I at both dihedral settings and $\delta = 20, 30^\circ$. Actual triangle area supplies the denominator. Circles/solid: $\epsilon = 0$; squares/dashed: $\epsilon = 10^{-5}$. Open diamonds identify symmetry-reflected zero-incidence solutions. All selected samples, including the nonmonotonic inclined 4° states, are shown. The indicator measures reversed chordwise wall shear over the specified region.

samples are nonmonotonic: between 2 and 4° , the reversal fraction falls from 0.579 to 0.090 at $\delta = 20^\circ$, and from 0.912 to 0.203 at $\delta = 30^\circ$. Their flow-state origin remains undetermined by the available steady samples.

The inclined $\delta = 30^\circ, \alpha = 2^\circ$ solution also contains reversed chordwise shear in the upper outboard leading-edge region ($2/3 \leq \eta \leq 0.95, 0 \leq \xi \leq 0.15$). Modeled BCM intermittency on the reversing-centroid subset has median 2.4×10^{-5} and maximum 8.6×10^{-5} , with zero wall eddy viscosity and SA working variable. These wall-model quantities characterize the stored local state; a laminar/turbulent boundary-layer profile or laminar separation bubble cannot be identified from them.

V. Dihedral and Planform Dependence

A. Matched dihedral: levels and incidence changes

The dihedral comparison holds Geom. I, δ , α , S , and MAC fixed. Orientation and the associated root cut/domain change together. The two settings support a controlled comparison within this dataset, without interpolation to smaller dihedral angles or generalization to aircraft tails.

Measured $\delta = 20^\circ$ absolute lift departures are smaller with dihedral, but their relative suppression depends on the incidence interval. Over $-2 \rightarrow 0^\circ$, the descriptive deficits relative to each preceding fitted increment are 80.6% and 76.4% for $\Gamma = 0$ and 45° ; over $-2 \rightarrow 2^\circ$, they are 64.5% and 82.2%. These ratios describe sampled contrasts without an assigned significance level. Lower absolute lift and weaker relative suppression are different statements.

Numerical projection accounting provides context for the absolute level difference. At $\alpha = 0^\circ$, $\delta = 20^\circ$, projection of the zero-dihedral normal load contributes about -0.247 of the matched lift difference -0.301 . Changes in local normal loading and span-force projection complete the balance. This coordinate decomposition describes the level difference; it does not isolate a causal contribution to nonlinear lift growth.

For the incidence-dependent moment comparison, Fig. 9 instead evaluates

$$\mathcal{I}_M = [C_{M,45}^{A_I}(\alpha_b) - C_{M,45}^{A_I}(\alpha_a)] - [C_{M,0}^{A_I}(\alpha_b) - C_{M,0}^{A_I}(\alpha_a)] \quad (6)$$

with identical incidence intervals and the common numerical axis. The corresponding contrasts of the integrated $z dF_x$ and $-(x - x_{A_I})dF_z$ terms complete its sum. For $\delta = 30^\circ$ over $-2 \rightarrow 0^\circ$, $\mathcal{I}_M = -0.0184$, with both terms negative. Their changing contributions across the other intervals show how geometric moment weighting enters the incidence-dependent response. These are ordinary interval contrasts without baseline subtraction.

B. Second planform and scope of the result

For $\text{II}/\delta = 20^\circ$, Table 5 again shows positive ordinary fixed lift growth and negative control growth, with negative departures of both components. Its fitted-center moment departures are both positive and reinforce. A small partial cancellation about the original numerical origin is reference-specific and sensitive to baseline and choice of zero-incidence solution (Appendix A).

The second-planform result accompanies the poorly reproduced measured lift slowdown described in Sec. III. The direct and symmetry-reflected zero-incidence solutions and their corresponding coefficients remain distinct in Fig. 4, with their interval sensitivity in the appendix. Since sweep, taper, aspect ratio, chord distribution, area, and MAC Reynolds number differ between I and II, the comparison describes the two tested planforms rather than an isolated sweep effect.

These results distinguish continuing integrated lift growth from reduced growth of its component contributions

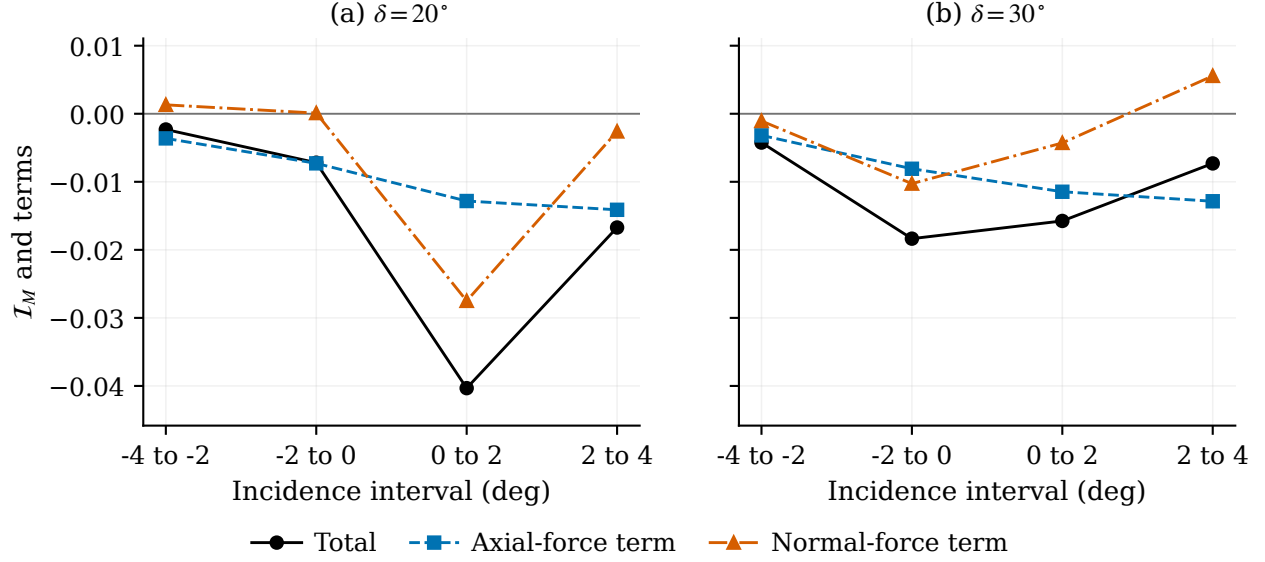


Fig. 9 Matched I dihedral contrasts of ordinary moment increments, 45° minus 0° , at $\delta = 20^\circ$ (a) and 30° (b). Moment uses common numerical axis $A_I = (0.642, 0, 0)$ m. The axial- and normal-force series are interval contrasts of the two normalized integrated terms in Eq. 5, whose sum is I_M in Eq. 6. Each marker represents its labeled 2° interval; lines guide the eye between interval summaries. Direct zero-incidence solutions are used, without baseline subtraction.

at fixed deflection. Moment adds the spatial weighting of those changes. A lift-only description or a model with constant slopes consequently loses information about load allocation and pitching response. The present departures vary incidence, not control deflection, and do not measure a control derivative. Their engineering interpretation is confined to the tested planforms, full-span control geometry, and conditions.

VI. Conclusions

The measured pre-stall response contains distinct lift and moment changes. The baseline 20° lift slowdown is pronounced, while a moment-response change is present at 10° . Inclined weak moment features and the curved early 30° response require separate incidence and uncertainty descriptions.

For $I/\Gamma = 0^\circ/\delta = 20^\circ$, the numerical fixed surface continues gaining lift while both fixed and control contributions fall below their preceding trends. About 63% of the total lift departure is allocated to the fixed part across the tested windows and zero-incidence solutions; the absolute departure is more sensitive to the baseline. Sectional pressures and spatial load sums show increased forward suction together with hinge/aft losses and reduced expected growth in both fixed-chord regions. The fixed and control moment departures about the fitted aerodynamic center reinforce, with the control contributing more strongly than the fixed surface. Similar aft force losses also produce larger moment changes outboard.

Matched dihedral comparisons separate absolute load projection from relative lift suppression and quantify the

incidence-dependent contributions of both moment-arm terms. Both fitted-center moment departures also reinforce for the second planform. These findings describe the tested configurations; their simultaneous geometric differences limit single-parameter generalization.

The detailed load accounting applies to the computed states. Partial WT–RANS correspondence includes substantial baseline lift bias, inclined irregularities, and an inadequately reproduced second-planform lift slowdown. The existing 35° mesh study provides integrated-load sensitivity evidence at its tested conditions, while the present spatial interpretation remains bounded by those accuracy limits. The reference computed case demonstrates that continued lift growth can mask reduced growth of both surface contributions, whose different spatial weighting produces a distinct pitching-moment response.

A. Sensitivity of the Numerical Loading Interpretation

Table 6 keeps the absolute baseline departure and its allocation separate. The three unweighted windows use the same samples for all components. The direct and symmetry-reflected zero-incidence solutions are not averaged. The earliest pre-feature window has appreciable curvature; the positive fitted-center moment departure and fixed lift fraction persist even though the total lift deficit varies.

Table 6 $I/\Gamma = 0^\circ/\delta = 20^\circ$, $0 \rightarrow 2^\circ$: **baseline and zero-incidence sensitivity.** D_M^A uses $x_A = 0.642$ m. **Fixed fraction is $D_{L,\text{fixed}}/D_{L,\text{total}}$, without an assigned confidence level.**

Zero solution	Baseline (deg)	Total D_L	Fixed fraction (%)	Total D_M^A
Direct	{−6, −4, −2}	−0.110605	62.55	+0.033238
Direct	{−8, −6, −4}	−0.136788	62.43	+0.040913
Direct	{−4, −2, 0}	−0.105014	62.64	+0.033958
Reflected	{−6, −4, −2}	−0.111790	62.76	+0.033493
Reflected	{−8, −6, −4}	−0.137973	62.60	+0.041168
Reflected	{−4, −2, 0}	−0.106792	62.98	+0.034341

The whole fore/aft sums survive the tested span-third versus span-quarter partition; finer chord bins preserve the two negative fixed-chord departures. Redefining the inboard aft boundary from $\eta < 1/3$ to $\eta < 1/4$ changes its lift departure from −0.01270 to −0.00939. This is partition sensitivity, not a regional discretization-error estimate. Exact matching to the solver surface groups is available for six direct baseline states, with complete facet coverage; other spatial results use geometric bins explicitly.

The $I/\delta = 20^\circ$ upper-aft reversal fraction at direct zero is 0.14578 versus 0.14089 at the reflected zero. Threshold 10^{-5} changes the direct $0 \rightarrow 2^\circ$ sequence to 0.14497–0.48043, preserving growth. Typical stored vector skin-friction norms are of order 10^{-3} ; the small thresholds test near-zero sensitivity, without a statistical uncertainty interpretation.

At the original numerical origin, the baseline total moment departure is +0.10028, with fixed +0.05286 and control +0.04743. At the fitted aerodynamic center the total is +0.03324, fixed +0.01069, and control +0.02255. This reference

dependence changes allocation while retaining reinforcement. All reference transports use body $C_{F,z}$ in Eq. 2, rather than lift. The principal experimental comparison remains at its own fitted center and is not reassigned to either numerical axis.

A. Saved-state and reference sensitivity

Terminal histories match the force reports for 41 of the 44 interpreted surface states. The exceptions are $I/\Gamma = 0^\circ/\delta = 30^\circ$ at displayed $\alpha = -6^\circ$, and $II/\delta = 20^\circ$ at displayed $\alpha = -4^\circ$ and direct 0° . Complete final-stage reconstruction-order registration is unavailable for every state. Terminal consistency and small iterative ranges do not establish steady-flow uniqueness or physical accuracy.

The early $I/\delta = 30^\circ$ fitted-center total moment departure varies from approximately -0.0262 to $+0.0076$ across the tested windows. For $II/\delta = 20^\circ$, the symmetry-reflected zero-incidence solution gives $0 \rightarrow 2^\circ$ lift increment 0.1121 instead of direct 0.0935 and fitted-center moment departure 0.0070 instead of 0.0146 . Each coefficient and surface field retains its own saved state; these alternate zero states are not a hysteresis measurement. About the original numerical origin, the direct II fixed/control moment departures are $-0.00231/+0.00352$, with total $+0.00121$. This small cancellation is reference-specific and does not describe the fitted-center reinforcement.

At the original numerical origin, the baseline aft-span moment increments are approximately $0.0082, 0.0129, 0.0156$ from inboard to outboard. Their magnitudes differ from those at the fitted center, while the outboard weighting persists.

Acknowledgments

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